

COVER PAGE

IFB - Interface Board

Schematic Version: 1.1.0

		
Project: BEE-DAL	Sub-system: OBC	
Design by: Cam Doan	Sheet title: CoverPage.SchDoc	Sheet 1 of 34
Checked by: Bao Bui	Document No: 1	Size: A3
Approved by: Vu Pham	Revision: 1.1.0	Date: 09/09/2026

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NOTE:

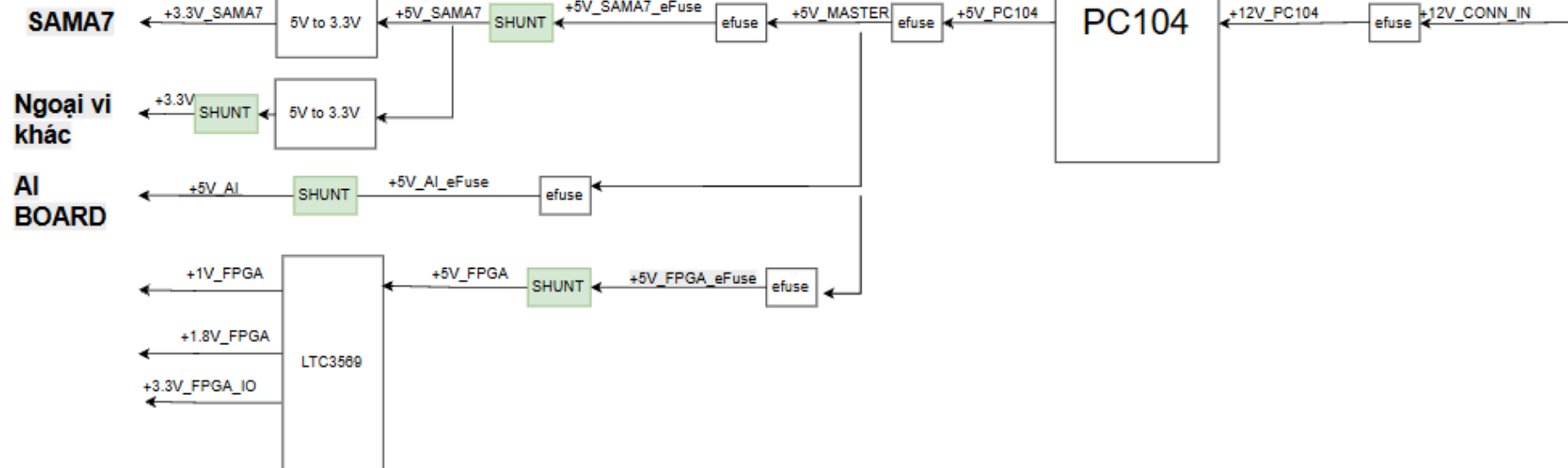
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- V1.0.0

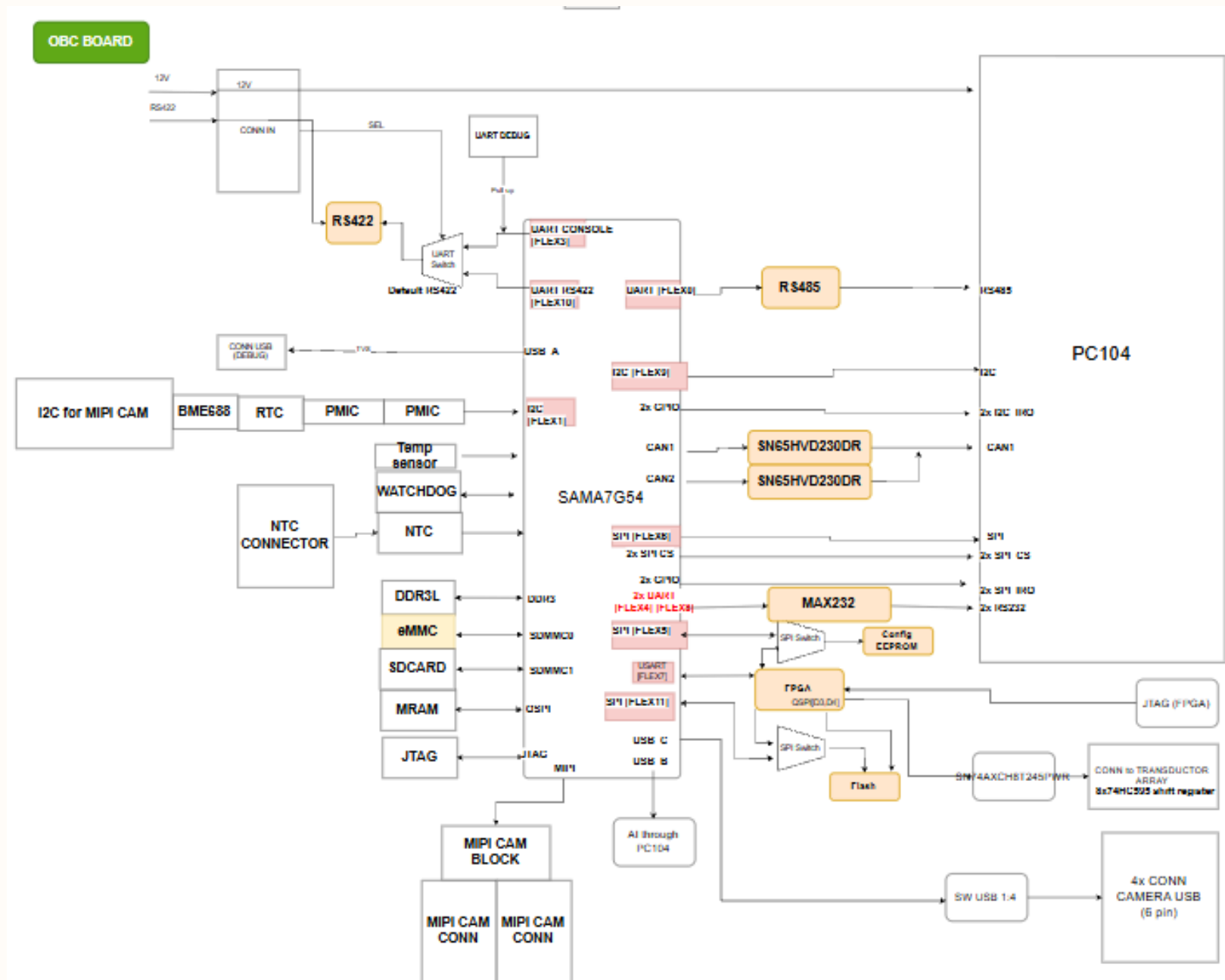
- V1.1.0: Add FPGA - Change to SAMA7G54



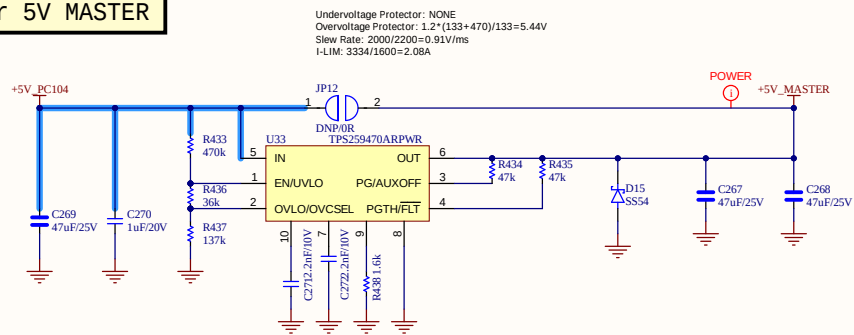
Project: BEE-DAL	Sub-system: OBC	
Design by: Cam Doan	Sheet title: Index.SchDoc	Sheet 2 of 34
Checked by: Bao Bui	Document No: 2	Size: A3
Approved by: Vu Pham	Revision: 1.1.0	Date: 09/09/2026



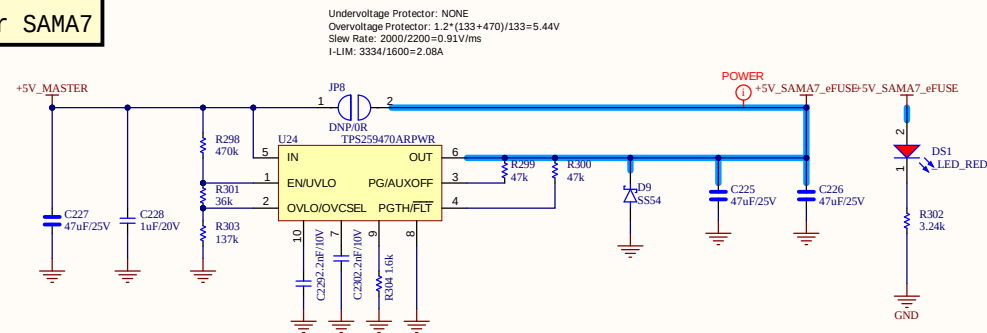
Project: BEE-DAL	Sub-system: OBC	
Design by: Cam Doan	Sheet title: PowerTree.SchDoc	Sheet 3 of 34
Checked by: Bao Bui	Document No: 3	Size: A3
Approved by: Vu Pham	Revision: 1.1.0	Date: 09/09/2026



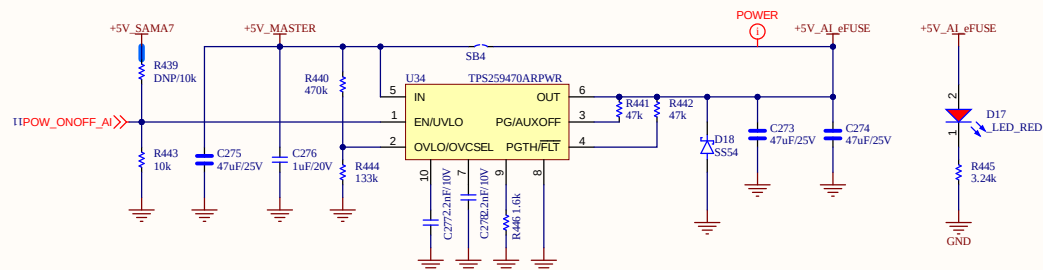
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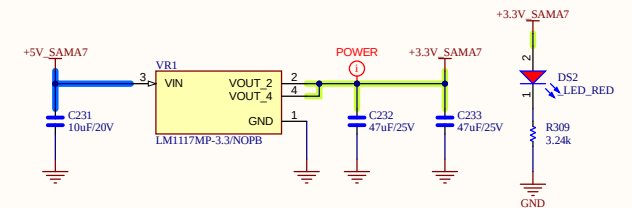
EFUSE for SAMA7



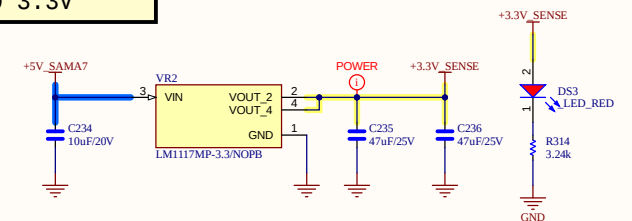
EFUSE for 5V AI



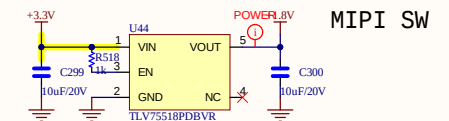
LD0 for SAMA7



LDO 3.3V



LD0 for MIPI SWITCH



Project: BEE-DAL

Sub-system: OBC

Design by: **Cam Doan**

Sheet title: Power.SchDoc

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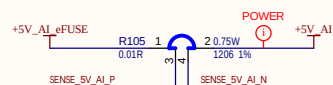
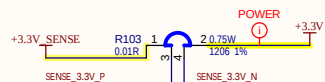
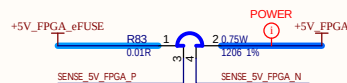
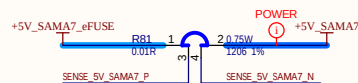
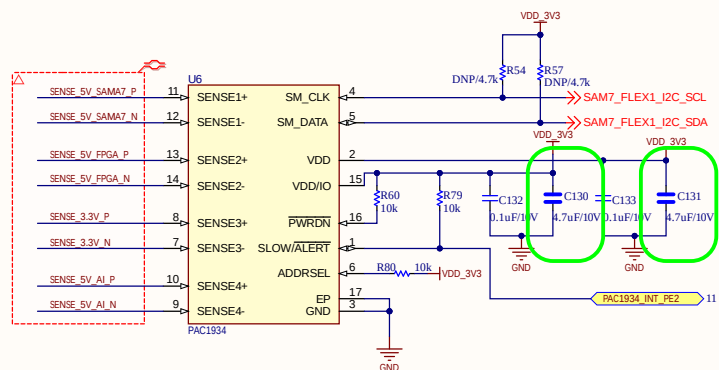
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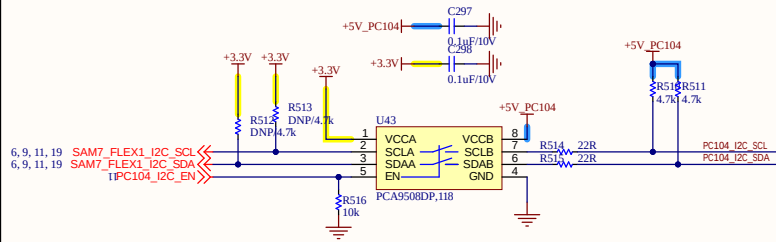
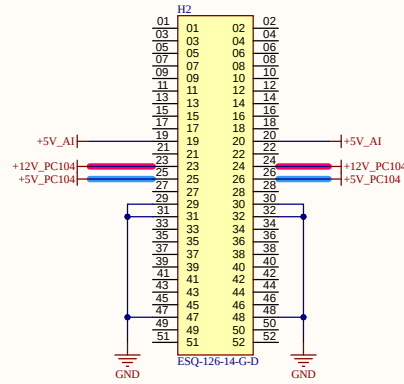
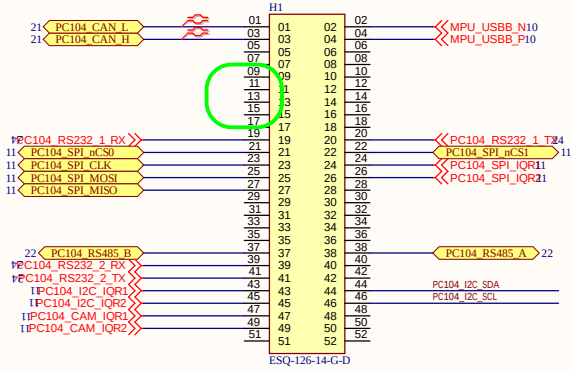
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POWER MEASUREMENT 12V AND 5V FPGA



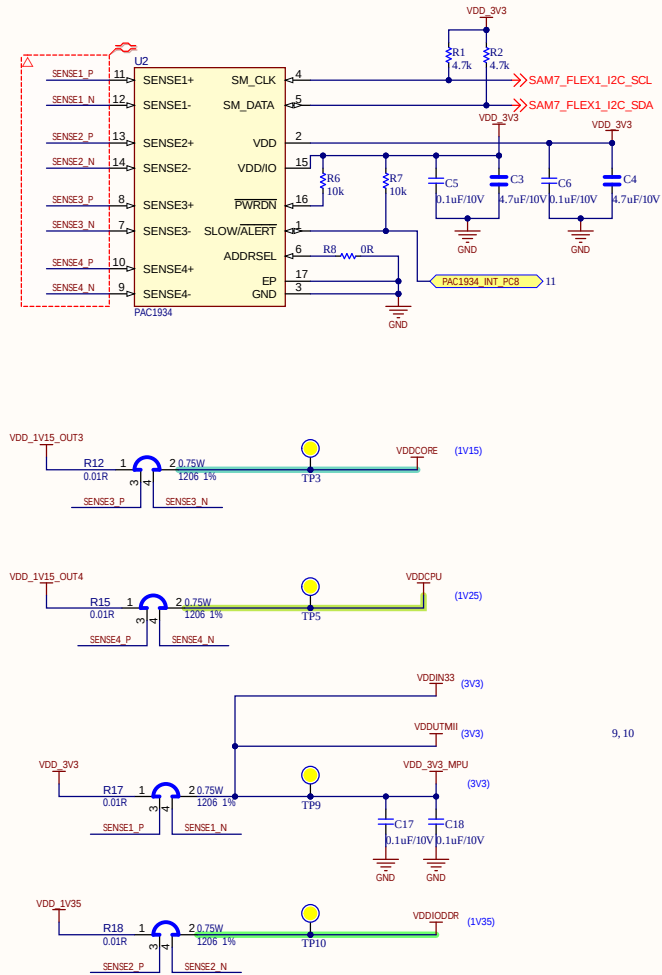
Project: BEE-DAL	Sub-system: OBC	
Design by: Cam Doan	Sheet title: POWER_MANAGER.SchDoc	Sheet 6 of 34
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Approved by: Vu Pham	Revision: 1.1.0	Date: 09/09/2026

PC104

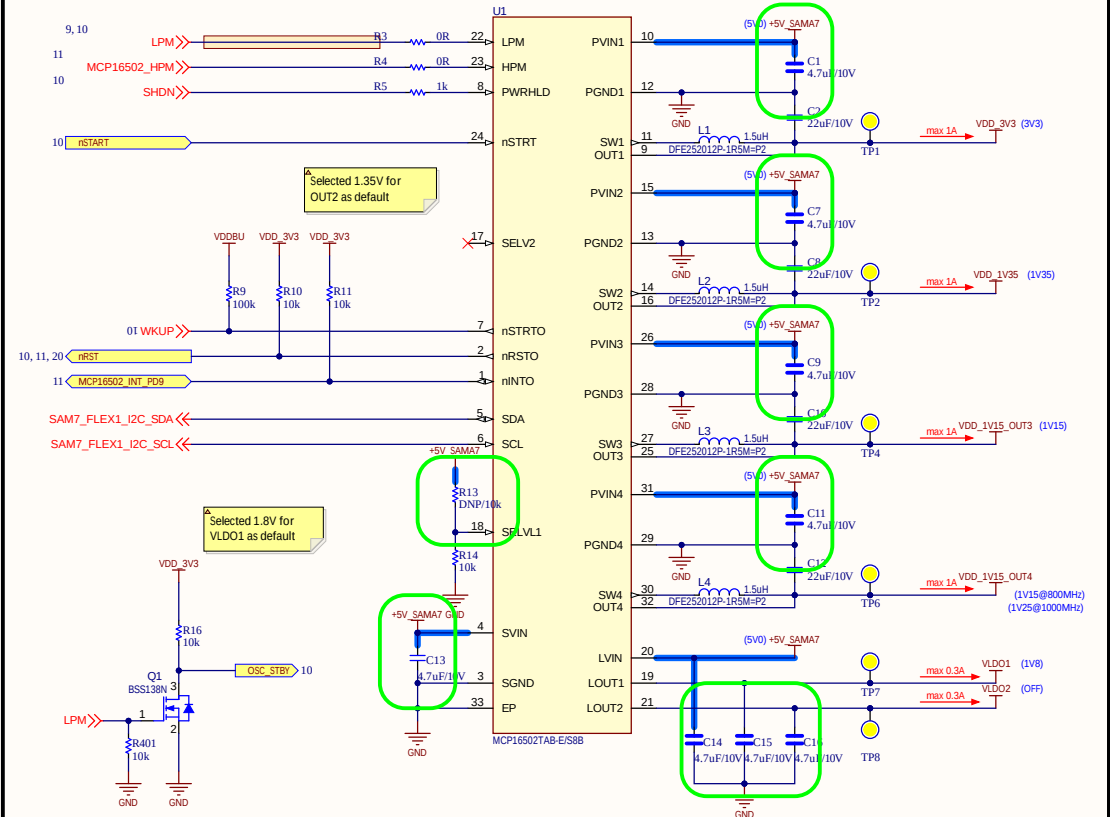


Project: BEE-DAL	Sub-system: OBC	
Design by: Cam Doan	Sheet title: CONNECTOR_BUS.SchDoc	Sheet 8 of 34
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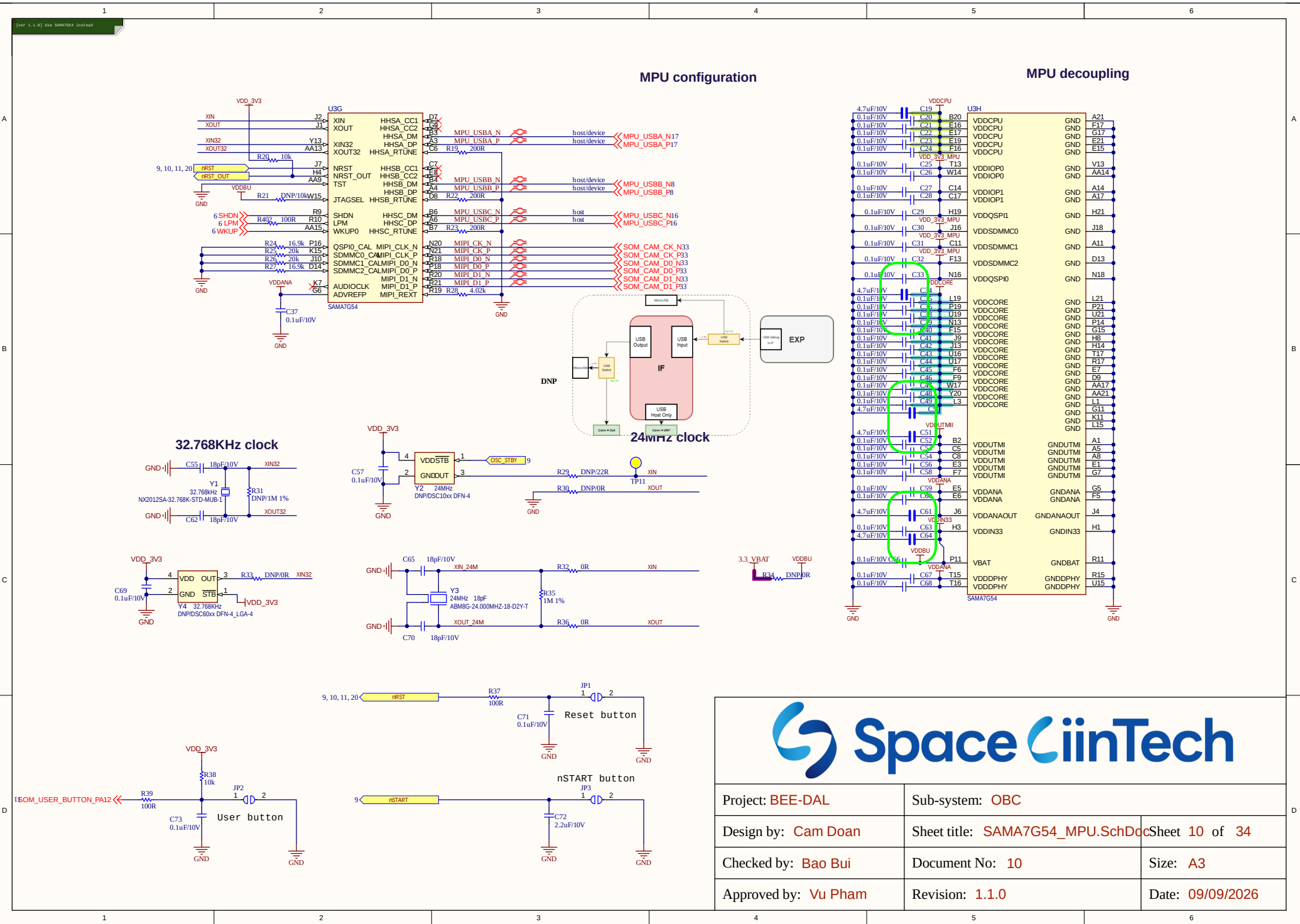
Power Measurement and Distribution



PMIC

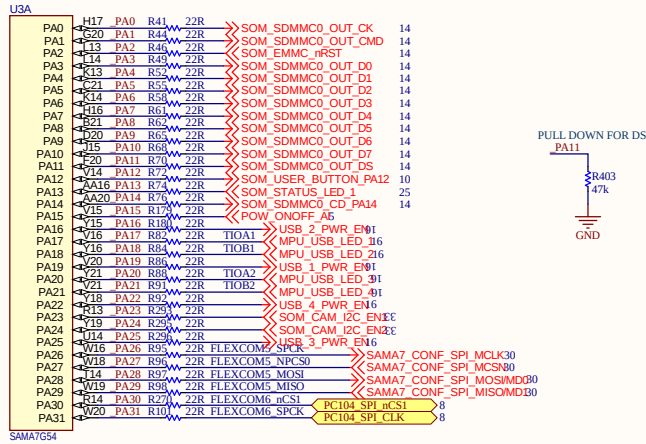


Project: BEE-DAL	Sub-system: OBC
Design by: Cam Doan	Sheet title: SAMA7G54_PMIC.SchDoc
Checked by: Bao Bui	Document No: 9
Approved by: Vu Pham	Revision: 1.1.0
	Size: A3
	Date: 09/09/2026

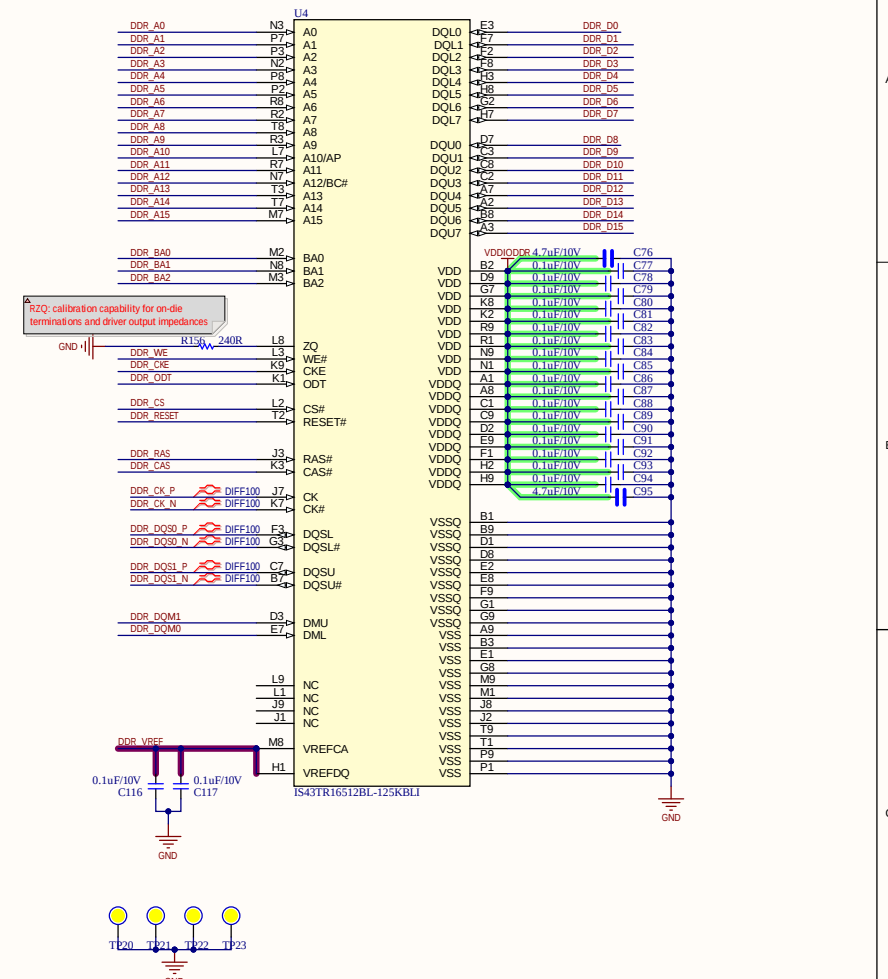
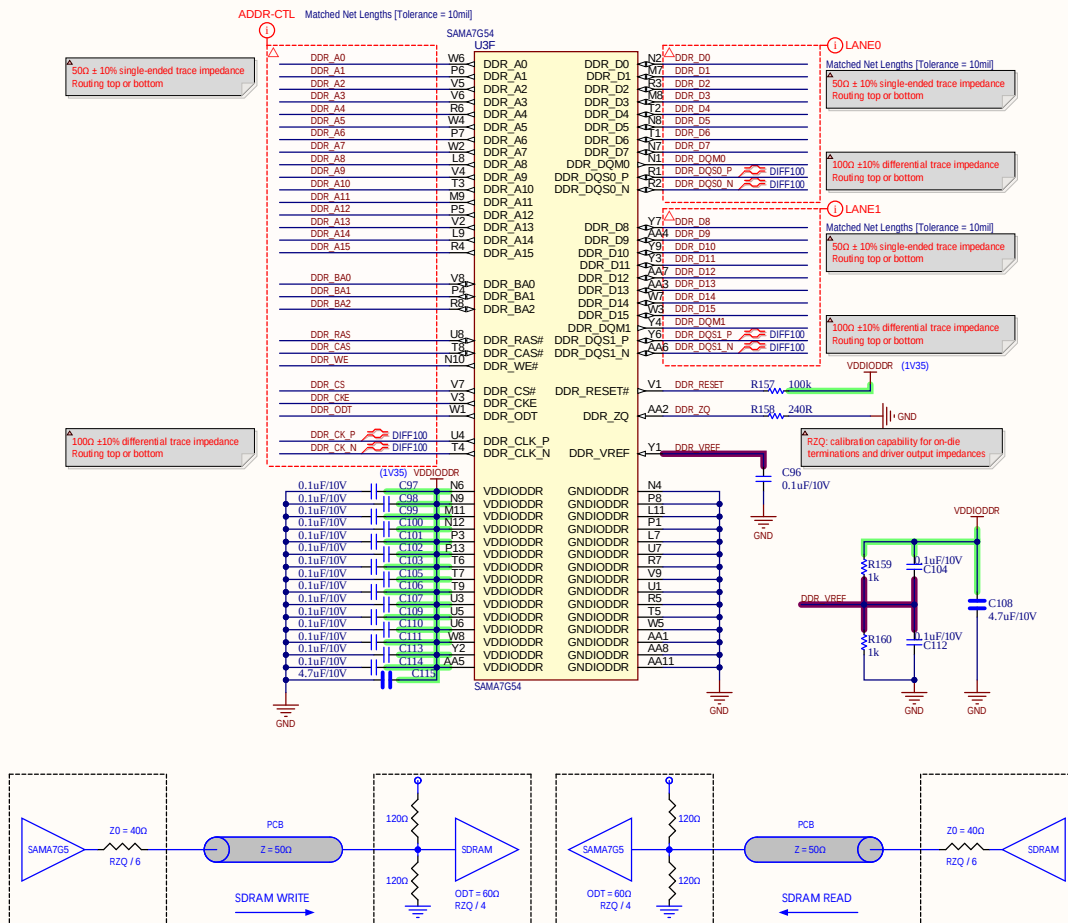


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SAMA7G5 Bus Distribution

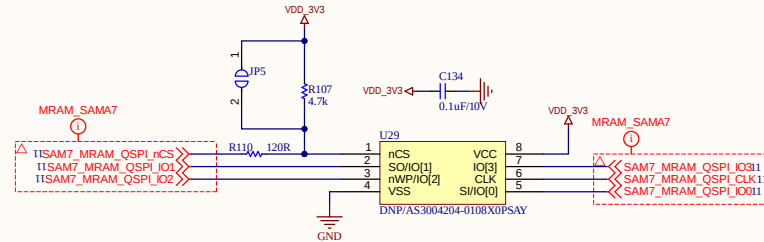


SAMA7G5 DDR3-1066 controller

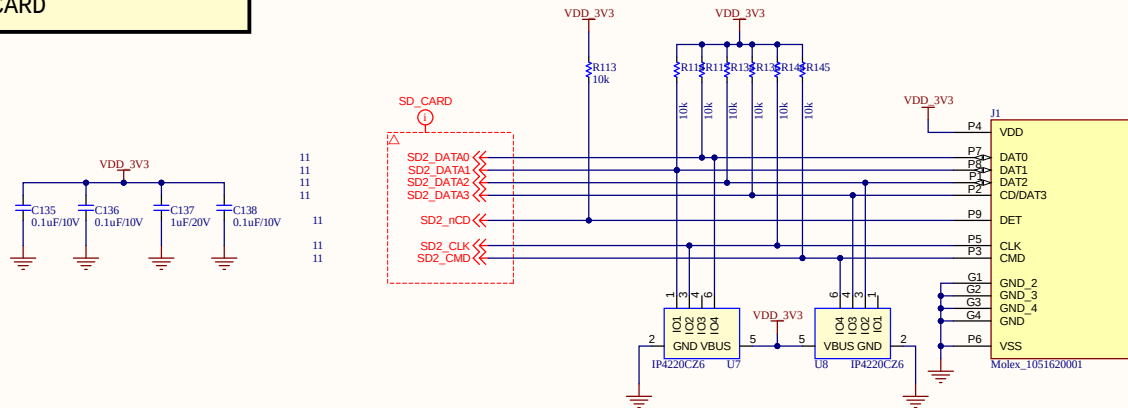


Project: BEE-DAL	Sub-system: OBC
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MRAM 4Mbit



SD CARD



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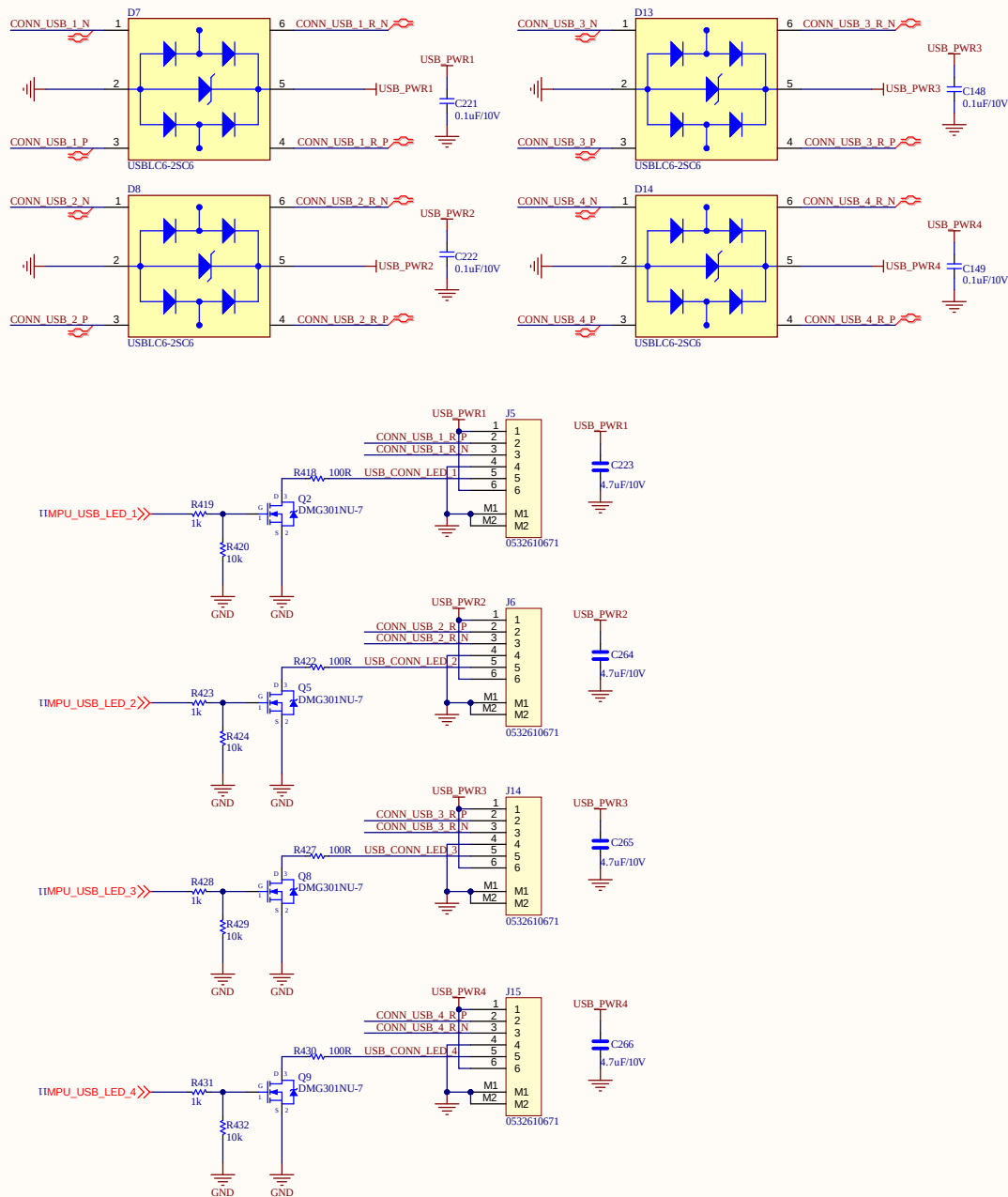
[illegible]

1	2	3	4	5	6
A					
B					
C					
D					
1	2	3	4	5	6

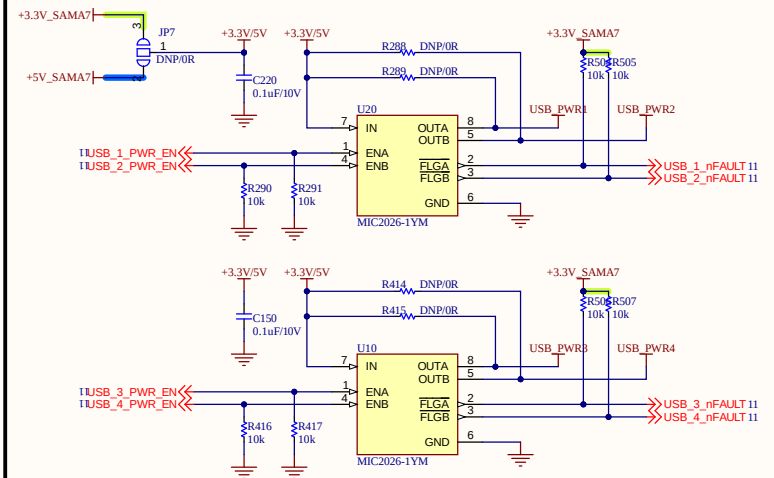


Project: BEE-DAL	Sub-system: OBC	
Design by: Cam Doan	Sheet title: USB_CAMERA_2.SchDocSheet 15 of 34	
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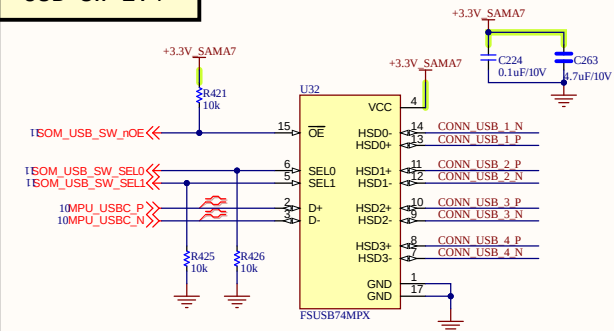
USB Camera



POWER SWITCH



USB SW 1:4



dùng N mos đk
2 mipi cam
3 usb cam
usb len AI board
eFuse cho 5V board AI t



Project: BEE-DAL	Sub-system: OBC	
Design by: Cam Doan	Sheet title: USB_CAMERA.SchDoc	Sheet 16 of 34
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UART MUX 4:1

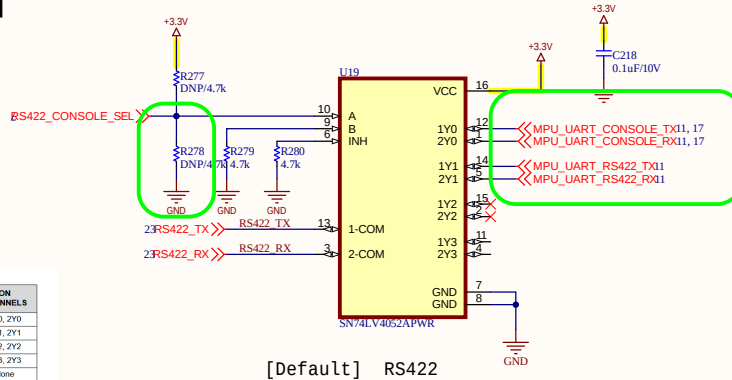
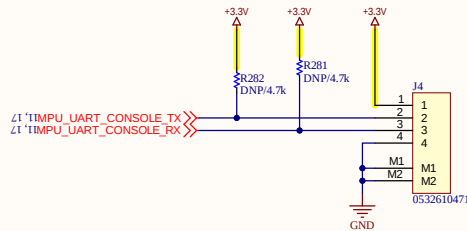


Table 7-1: Function Table

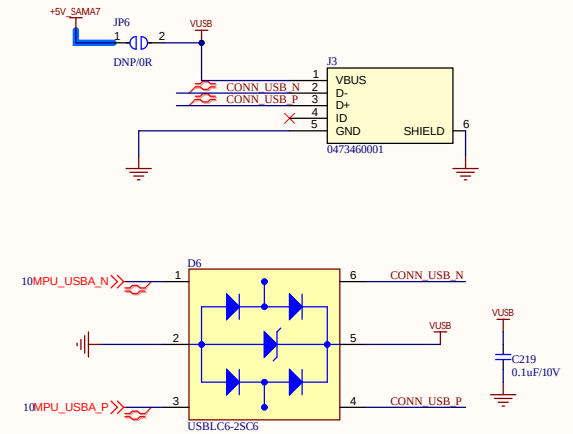
INPUTS			ON CHANNELS
INH	B	A	
L	L	L	1Y0, 2Y0
L	L	H	1Y1, 2Y1
L	H	L	1Y2, 2Y2
L	H	H	1Y3, 2Y3
H	X	X	None

[Default] RS422

UART CONSOLE

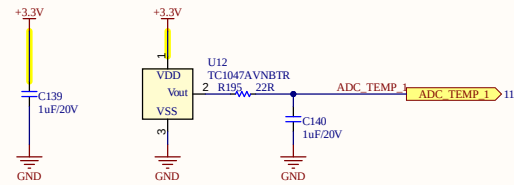


USB CONSOLE



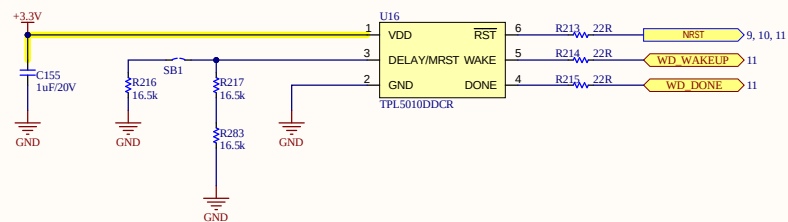
Project: BEE-DAL	Sub-system: OBC	
Design by: Cam Doan	Sheet title: DEBUG_CONSOLE.SchDoc	Sheet 17 of 34
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SENSOR TEMP

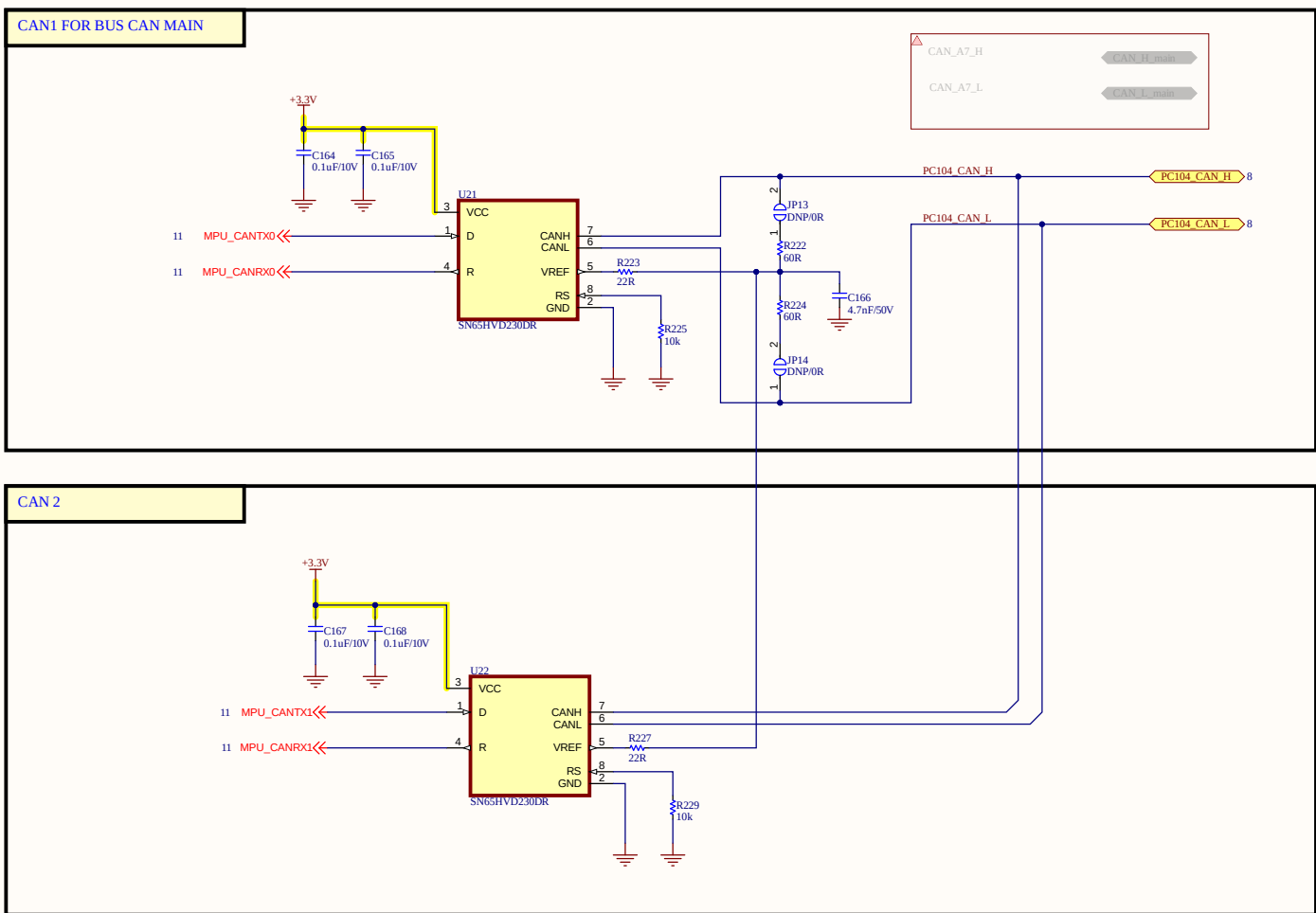


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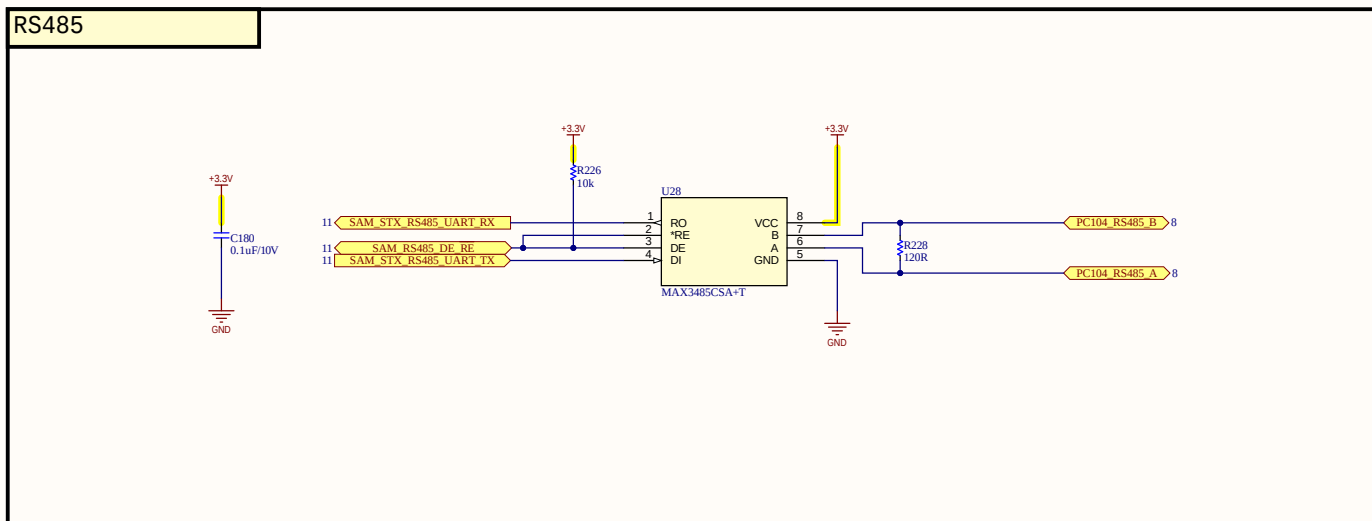
WATCHDOG



Project: BEE-DAL	Sub-system: OBC	
Design by: Cam Doan	Sheet title: WATCHDOG.SchDoc	Sheet 20 of 34
Checked by: Bao Bui	Document No: 20	Size: A3
Approved by: Vu Pham	Revision: 1.1.0	Date: 09/09/2026

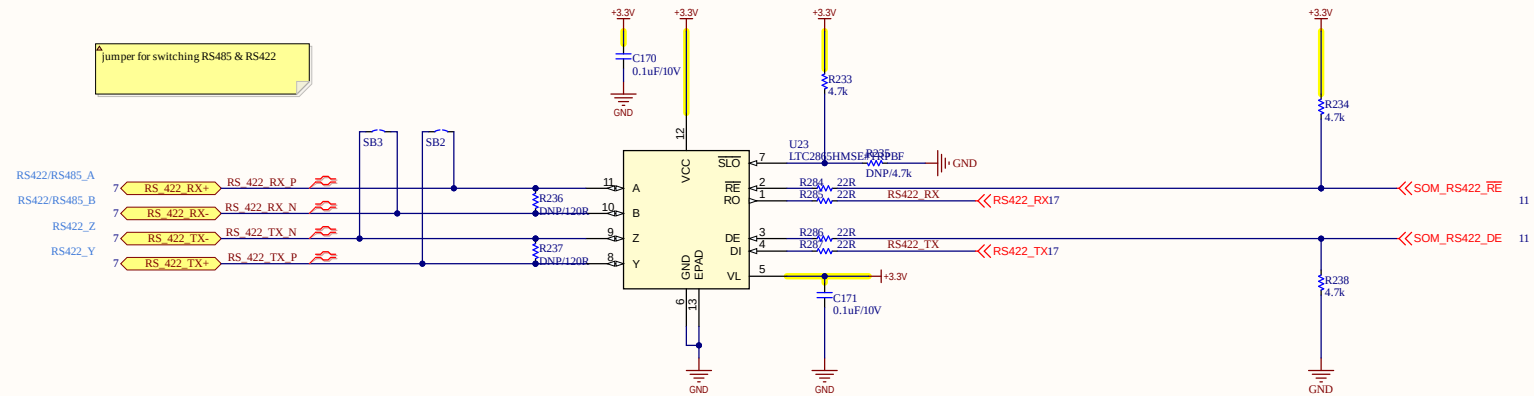


Project: BEE-DAL	Sub-system: OBC	
Design by: Cam Doan	Sheet title: CAN_PHY.SchDoc	Sheet 21 of 34
Checked by: Bao Bui	Document No: 21	Size: A3
Approved by: Vu Pham	Revision: 1.1.0	Date: 09/09/2026



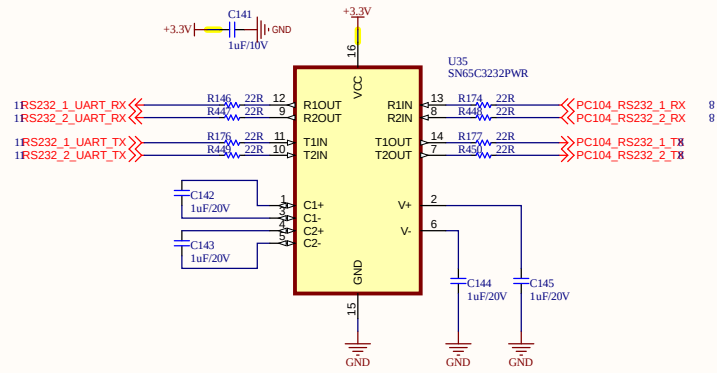
Project: BEE-DAL	Sub-system: OBC	
Design by: Cam Doan	Sheet title: RS485.SchDoc	Sheet 22 of 34
Checked by: Bao Bui	Document No: 22	Size: A3
Approved by: Vu Pham	Revision: 1.1.0	Date: 09/09/2026

RS422

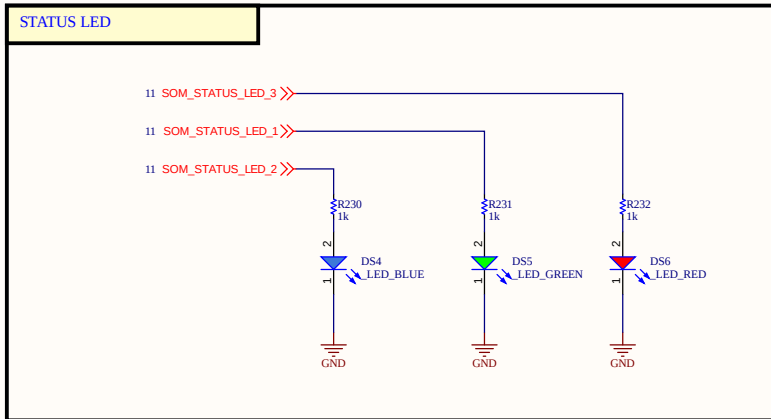


Project: BEE-DAL	Sub-system: OBC	
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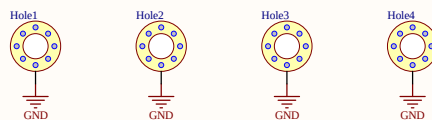
RS232



Project: BEE-DAL	Sub-system: OBC	
Design by: Cam Doan	Sheet title: RS232.SchDoc	Sheet 24 of 34
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Design by: Cam Doan	Sheet title: BUTTON_LED.SchDoc	Sheet 25 of 34
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Approved by: Vu Pham	Revision: 1.1.0	Date: 09/09/2026



Project: BEE-DAL

Sub-system: OBC

Design by: Cam Doan

Sheet title: MOUNTING_HOLE.SchDoc Sheet 26 of 34

Checked by: Bao Bui

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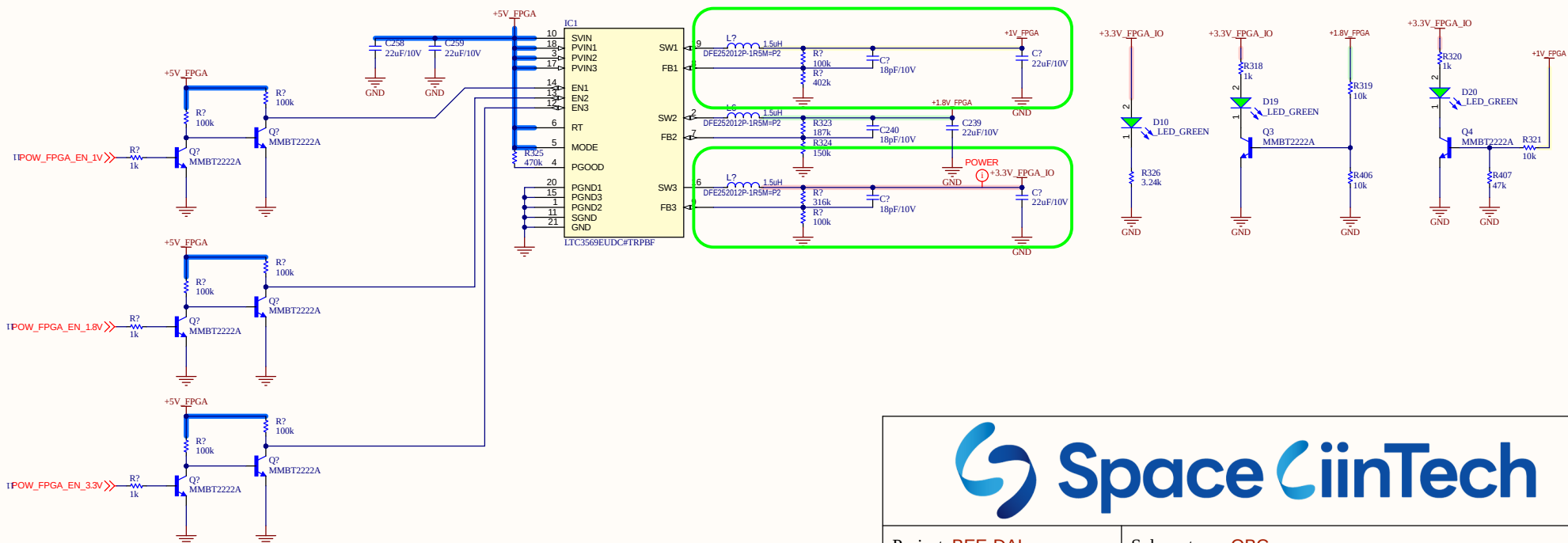
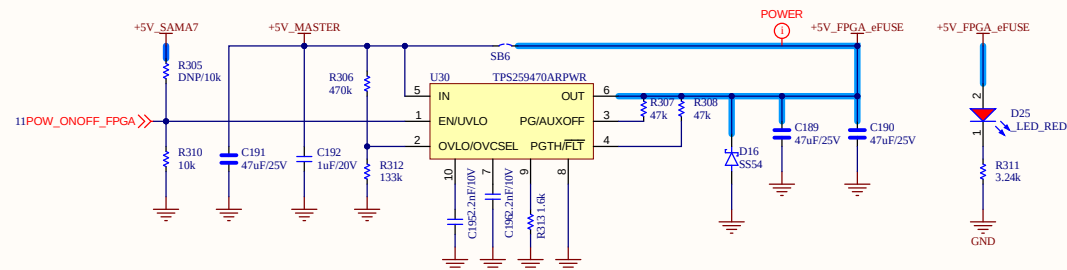
Size: A3

Approved by: **Vu Pham**

Revision: 1.1.0

Date: 09/09/2026

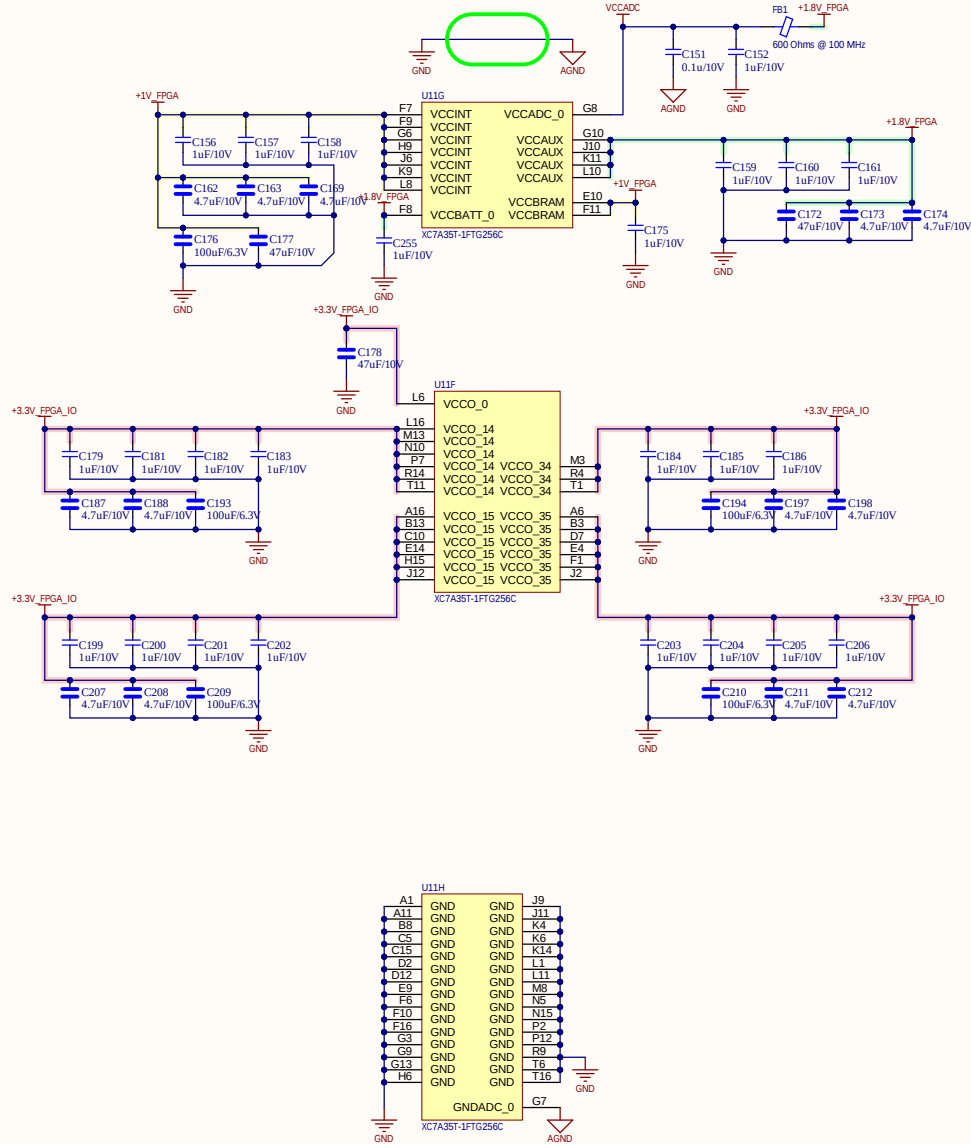
Main Source 5V



Project: BEE-DAL	Sub-system: OBC	
Design by: Cam Doan	Sheet title: FPGA_POWER.SchDoc	Sheet 27 of 34
Checked by: Bao Bui	Document No: 27	Size: A3
Approved by: Vu Pham	Revision: 1.1.0	Date: 09/09/2026

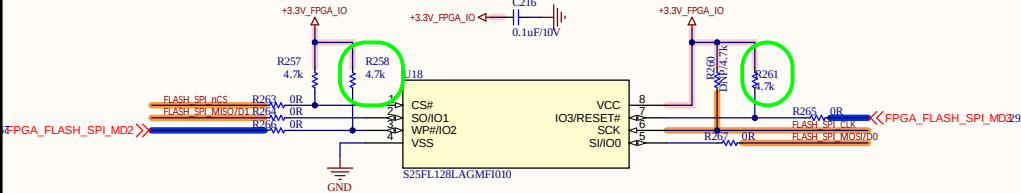
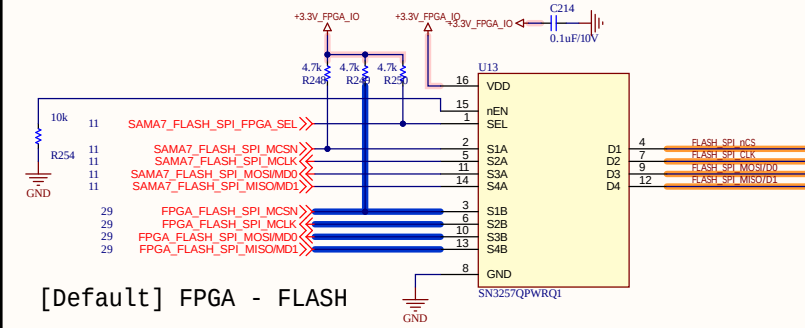
Table 3.1. VCCIO Supply Options

VCCIO Bank	Selection Jumper	3.3 V	1.8 V	1.5 V	1.2 V
VCCIO0	J47	Selectable	Default	—	—
VCCIO1	—	Fixed	—	—	—
VCCIO2	—	Fixed	—	—	—

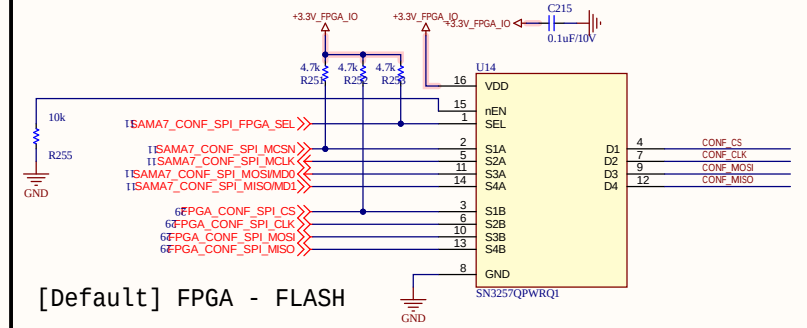


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Design by: Cam Doan	Sheet title: FPGA_POWER_2.SchDoc	Sheet 28 of 34
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FPGA FLASH



FPGA EEPROM



Project: BEE-DAL

Sub-system: OBC

Design by: Cam Doan

Sheet title: FPGA_FLASH_EEPROM, Sheet No 80 of 34

Checked by: Bao Bui

Document No: 30

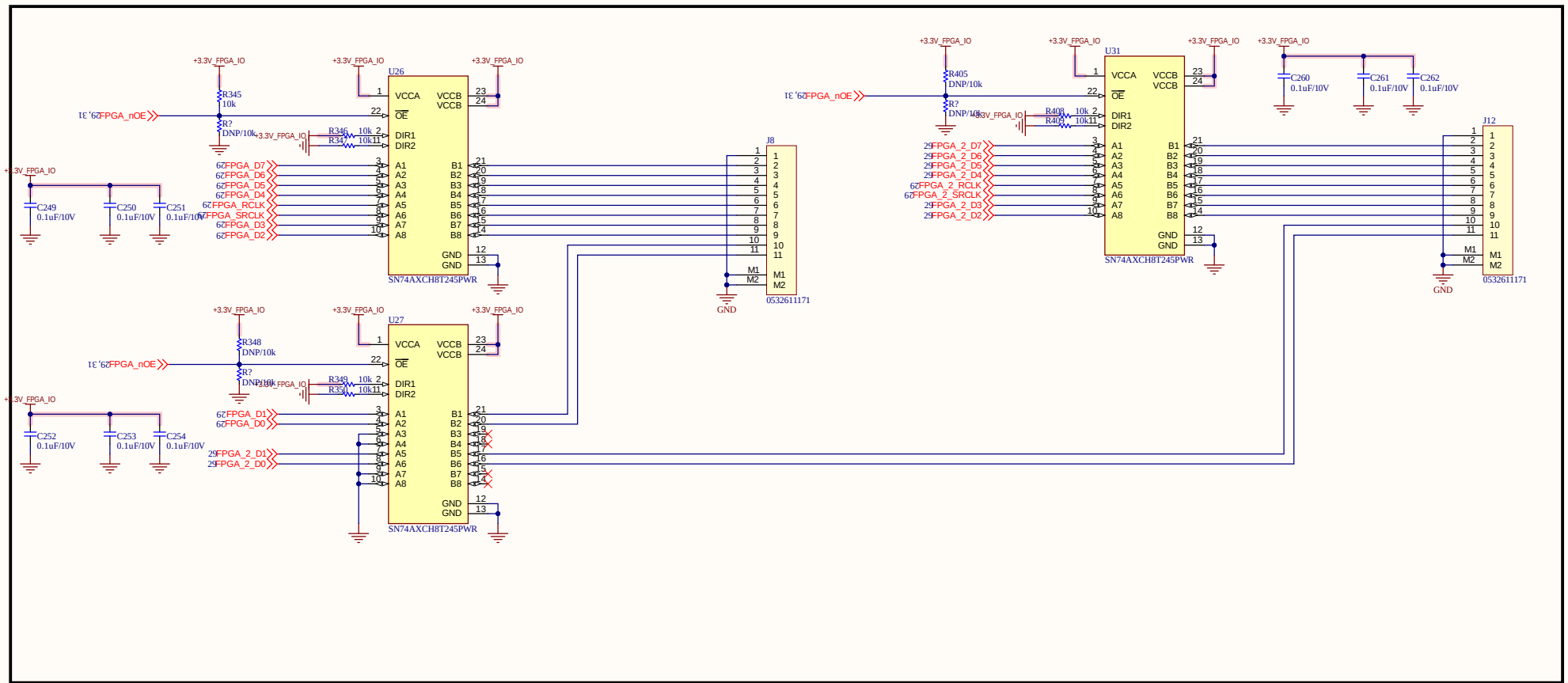
Size: A3

Approved by: Vu Pham

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L	L	H	B to A	A to B
L	H	L	A to B	B to A
L	H	H	A to B	B to A



Project: BEE-DAL	Sub-system: OBC
Design by: Cam Doan	Sheet title: FPGA_CONN_OUT.SchDoc Sheet 31 of 34
Checked by: Bao Bui	Document No: 31
Approved by: Vu Pham	Revision: 1.1.0
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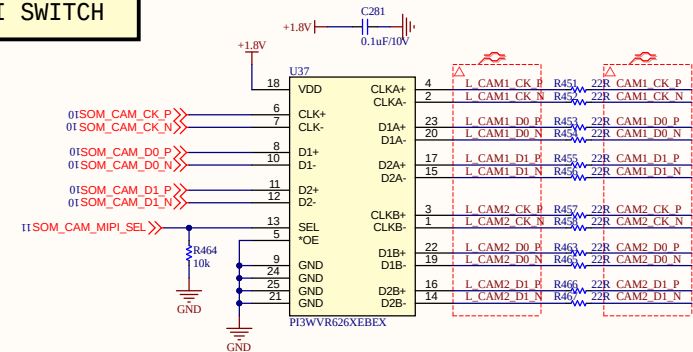
1	2	3	4	5	6	
A						A
B						B
C						C
D						D
1	2	3	4	5	6	

		
Project: BEE-DAL	Sub-system: OBC	
Design by: Cam Doan	Sheet title: ChangeNote.SchDoc	Sheet 32 of 34
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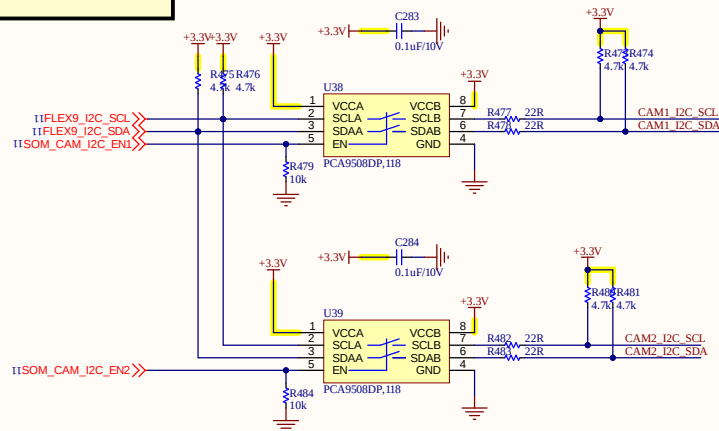


Project: BEE-DAL	Sub-system: OBC	
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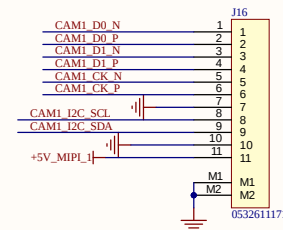
MIPI SWITCH



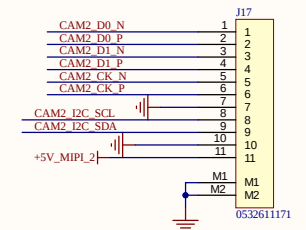
I2C SWITCH



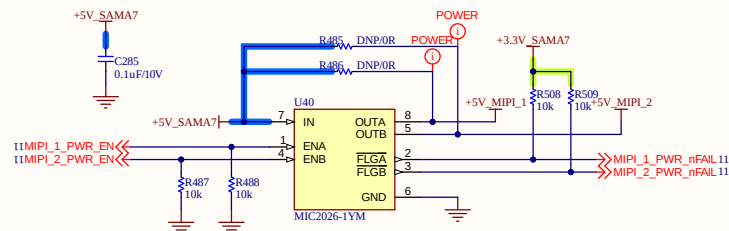
for CAMERA 1



for CAMERA 2



POWER SWITCH



Project: BEE-DAL

Sub-system: OBC

Design by: Cam Doan

Sheet title: MIPI_CSI.SchDoc

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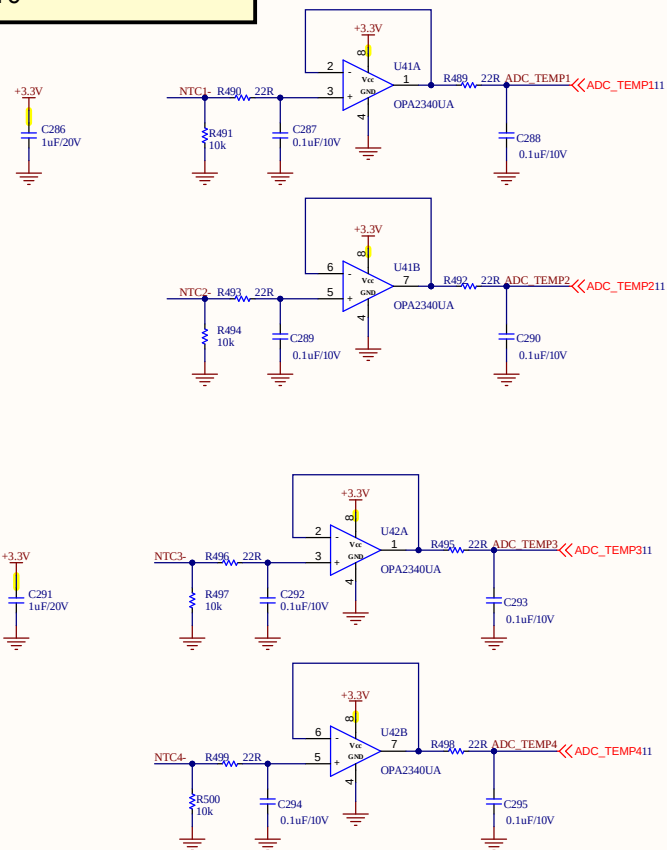
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Approved by: Vu Pham

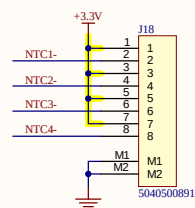
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NTC



CONNECTOR FOR NTC



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